

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							*****VALVES*****			812	NIPPLE TO BE CUT FROM SS SCH XXS PIPE.
20										816	COMPACT FLANGES CAN BE USED FOR TIE-IN PURPOSE ONLY TO EXISTING PIPING AND EQUIPMENT.
20							***** MODULAR VALVES *****				
30	U04		MODULAR DBB	K25RC02	0.5	1.5	MODULAR VALVE DB&B BALL TYPE 2500# RTJ LTCS ASTM A350 LF2 CL.1 INCONEL TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	817	USE ONLY FOR TIE IN
40	U04		MODULAR DBB	K25RC08	2	2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ LTCS ASTM A350 LF2 CL.1 INCONEL TRIM METAL GO - FULL SIZE TYPE 2BV	0	0	B1	ALL BOLTS ARE METRIC DIA.
50	U04		MODULAR DBB	M25RC02	0.75	1.5	MODULAR VALVE DB&B BALL TYPE 2500# RTJ LTCS ASTM A350 LF2 CL.1 INCONEL TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10
60	U04		MODULAR DBB	M25RC08	2	2	MODULAR VALVE DB&B BALL TYPE 2500# RTJ LTCS ASTM A350 LF2 CL.1 INCONEL TRIM METAL GO - CONTROLLED BORE TYPE 2BVR	0	0	B6	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22
70	U04	VAW	MODULAR DBB	M25HC02	0.75	1	MODULAR VALVE DB&B BALL TYPE 2500# RTJ LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR	2	2	FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.
80										SB6	USE JACKSCREWS FOR PADDLE
80		VB2					***** BALL VALVES *****				BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.
90										UJ2	USE 1 1/2IN WITH THERMOWELL
90	U04		BALL	A25RC07	2	2	BALL 2500# RTJ LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT RB GO	0	0		CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.
100	U04	817	BALL	A25CC07-K1	3	4	BALL 2500# CF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT RB GO	0	0		
110	U04		BALL	B25RC01	0.75	1.5	BALL 2500# RTJ LTCS ASTM A350 LF2 CL.1 INCONEL TRIM FLOATING SOFT FB LO	0	0		
120	U04	VA5	BALL	B25RC07	2	2	BALL 2500# RTJ LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT FB GO	0	0		
130	U04	817	BALL	B25CC07-K1	4	4	BALL 2500# CF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT FB GO	0	0	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
140										VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
140							***** GLOBE VALVES *****				
150	U04		GLOBE	L25RC48	0.75	1.5	GLOBE 2500# RTJ LTCS A350 LF2 CL.1 INCONEL OS&Y HW	0	0		
160	U04		GLOBE	L25RC41	2	2	GLOBE 2500# RTJ LTCS A352 LCC INCONEL OS&Y HW	0	0	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.
170	U04	817	GLOBE	L25CC42-K1	4	4	GLOBE 2500# CF LTCS A352 LCC INCONEL OS&Y GO	0	0		
180										VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.
180							***** CHECK VALVES *****				
190											
190	U04	VC1	CHECK	C25RC50	0.75	1.5	CHECK PISTON 2500# RTJ LTCS A350 LF2 CL.1 INCONEL TRIM	0	0		
200	U04		CHECK	C25RC52	2	2	CHECK DUAL 2500# RTJ LTCS A352 LCC TRIM 16	0	0		
210	U04		CHECK	C25RC45	2	2	CHECK DUAL WAFER LUG 2500# RTJ LTCS A352 LCC INCONEL TRIM THROU BOLTED	0	0		
220	U04	817	CHECK	C25CC52-K1	4	4	CHECK DUAL DBL FLG 2500# CF LTCS A352 LCC INCONEL TRIM	0	0		
230											
230							*****PIPES*****				
240											
240											
240		812	PIPE	5358511	0.75	1.5	PIPE SCH XXS SMLS LTCS A333 GR.6 PE	0	0		
250			PIPE	5358516	2	2	PIPE WT 12.0MM SMLS LTCS A333 GR.6 BE	0	0		
260			PIPE	5358536	3	3	PIPE WT 16.0MM SMLS LTCS A333 GR.6 BE	0	0		
270			PIPE	5358537	4	4	PIPE WT 20.0MM SMLS LTCS A333 GR.6 BE	0	0		
280	U04		PIPE	5358539	6	6	PIPE WT 29.0MM SMLS LTCS A333 GR.6 BE	0	0		
290											
290							***** FITTINGS *****				
300											
300			ELBOW	5392870	0.75	1.5	ELL 90 DEG LR XXS LTCS A420 WPL6	0	0		
310			ELBOW	5392871	2	2	ELL 90 DEG LR WT 12.0MM LTCS A420 WPL6	0	0		
320			ELBOW	5392818	3	3	ELL 90 DEG LR WT 16.0MM LTCS A420 WPL6	0	0		
330			ELBOW	5392876	0.75	1.5	ELL 45 DEG LR XXS LTCS A420 WPL6	0	0		
340			ELBOW	5392875	2	2	ELL 45 DEG LR WT 12.0MM LTCS A420 WPL6	0	0		
350			ELBOW	5392832	3	3	ELL 45 DEG LR WT 16.0MM LTCS A420 WPL6	0	0		
360											
360			TEE	5392879	0.75	1.5	TEE SCH XXS LTCS A420 WPL6	0	0		
370			TEE	5392889	2	2	TEE WT 12.0MM LTCS A420 WPL6	0	0		
380			TEE	5392837	3	3	TEE WT 16.0MM LTCS A420 WPL6	0	0		
390			TEE RED	5392877	0.75	1	TEE RED SCH XXS LTCS A420 WPL6	1	1.5		
400			TEE RED	5392887	1	1	TEE RED WT 12.0MM / SCH XXS LTCS A420 WPL6	2	2		
410			TEE RED	5392842	1.5	1.5	TEE RED WT 16.0MM/SCH XXS LTCS A420 WPL6	3	3		
420			TEE RED	5392843	2	2	TEE RED WT 16.0MM/12.0MM LTCS A420 WPL6	3	3		
430			TEE RED	5392844	2	2	TEE RED WT 20.0MM/12.0MM LTCS A420 WPL6	4	4		
440											
440	U04		CAP	5392880	0.75	1.5	CAP BW SCH XXS LTCS A420 WPL6	0	0		
450			CAP	5392883	2	2	CAP WT 12.0MM LTCS A420 WPL6	0	0		
460			CAP	5392846	3	3	CAP WT 16.0MM LTCS A420 WPL6	0	0		
470											
470			RED CONC	5392881	0.75	1	REDUCER CONC SCH XXS LTCS A420 WPL6	1	1.5		
480			RED CONC	5392884	0.75	1.5	REDUCER CONC WT 12.0MM / SCH XXS LTCS A420 WPL6	2	2		
490			RED CONC	5392859	1.5	1.5	REDUCER CONC WT 16.0MM/XXS LTCS A420 WPL6	3	3		
500			RED CONC	5392852	2	2	REDUCER CONC WT 20.0MM/12.0MM LTCS A420 WPL6	4	4		
510			RED CONC	5392853	3	3	REDUCER CONC WT 20.0MM/16.0MM LTCS A420 WPL6	4	4		
520			RED ECC	5392882	0.75	1	REDUCER ECC SCH XXS LTCS A420 WPL6	1	1.5		
530			RED ECC	5392885	0.75	1.5	REDUCER ECC WT 12.0MM / SCH XXS LTCS A420 WPL6	2	2		

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - ALL SIZES. SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE NACE MR0175 / ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 2500 RF/CF TEMPERATURE RANGE: -40 / 200°C	12/16/2024	BY OZAC	U04	Page 1 of 2	2500K1

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540			RED ECC	5392862	1.5	1.5	REDUCER ECC WT 16.0MM/XXS LTCS A420 WPL6	3	3	
550			RED ECC	5392861	2	2	REDUCER ECC WT 16.0MM/12.0MM LTCS A420 WPL6	3	3	
560			RED ECC	5392863	1.5	1.5	REDUCER ECC WT 20.0MM / SCH XXS LTCS A420 WPL6	4	4	
570			RED ECC	5392867	2	2	REDUCER ECC WT 20.0MM / 12.0MM LTCS A420 WPL6	4	4	
580			RED ECC	5392866	3	3	REDUCER ECC WT 20.0MM / 16.0MM LTCS A420 WPL6	4	4	
590	U04		RED ECC	5392864	3	3	REDUCER ECC WT 29.0MM / 16.0MM LTCS A420 WPL6	6	6	
600										
600							***** OLETS *****			
610										
610	U04		OLETS	5504285	0.75	1.5	NIPOFLANGE 2500# RTJ WT 12.0MM/SCH XXS LTCS A350 LF2 CL.1	2	2	
620	U04		OLETS	5504286	0.75	1.5	NIPOFLANGE 2500# RTJ WT 16.0MM/SCH XXS LTCS A350 LF2 CL.1	3	3	
630	U04		OLETS	5504287	0.75	1.5	NIPOFLANGE 2500# RTJ WT 20.0MM / SCH XXS LTCS A350 LF2 CL.1	4	4	
640	U04	UJ2	OLETS	5504290	0.75	1.5	NIPOFLANGE 2500# RTJ WT 12.0MM/SCH XXS LTCS A350 LF2 CL.1	2	2	
650	U04	UJ2	OLETS	5504291	0.75	1.5	NIPOFLANGE 2500# RTJ WT 16.0MM/SCH XXS LTCS A350 LF2 CL.1	3	3	
660	U04	UJ2	OLETS	5504292	0.75	1.5	NIPOFLANGE 2500# RTJ WT 20.0MM / SCH XXS LTCS A350 LF2 CL.1	4	4	
670										
670	U04		OLETS	5505158	0.75	1	WELDOLET WT 12.0MM/SCH XXS LTCS A350 LF2 CL.1	2	2	
680	U04		OLETS	5505159	0.75	1	WELDOLET WT 16.0MM/SCH XXS LTCS A350 LF2 CL.1	3	3	
690	U04		OLETS	5505160	0.75	1	WELDOLET WT 20.0 MM / SCH XXS LTCS A350 LF2 CL.1	4	4	
700										
700	U04	UJ2	ELBOFLG	5516025	1.5	1.5	ELBOFLANGE 2500# RTJ WT 12.0MM/SCH XXS LTCS A350 LF2 CL.1	2	2	
710	U04	UJ2	ELBOFLG	5516026	1.5	1.5	ELBOFLANGE 2500# RTJ WT 16.0MM/SCH XXS LTCS A350 LF2 CL.1	3	3	
720	U04	UJ2	ELBOFLG	5516027	1.5	1.5	ELBOFLANGE 2500# RTJ WT 20.0MM /SCH XXS LTCS A350 LF2 CL.1	4	4	
730										
730		816					***** FLANGES *****			
740										
740	U04		FLANGE	5557706	0.75	1.5	FLG WN 2500# RTJ LTCS A350 LF2 CL.1 SCH XXS	0	0	
750	U04		FLANGE	5619890	2	2	FLG WN 2500# RTJ LTCS A350 LF2 CL.1 WT 12.0MM	0	0	
760	U04		FLANGE	5557714	3	3	FLG WN 2500# RF LTCS A350 LF2 CL.1 WT 16.0MM	0	0	
770	U04		FLANGE	5557715	4	4	FLG WN 2500# RF LTCS A350 LF2 CL.1 WT 20.0MM	0	0	
780	U04	FA8	TEE RED	5599713	0.5	0.75	FLG SCRD (FNPT) REDUCING 2500# RTJ LTCS A350 LF2 CL.1	0.75	1.5	
790										
790			FLANGE	5851202	0.75	0.75	FLANGE ADAPTER 2500# RTJ 316 SS X 12 MM TUBE	0	0	
800			FLANGE	5851202	2	2	FLANGE ADAPTER 2500# RTJ 316 SS X 12 MM TUBE	0	0	
810										
810	U04	817	FLANGE	5619902	4	4	FLG COMPACT 2500# RJ LTCS A350 LF2 CL.1 WT 20.0MM	0	0	
820	U04		BLIND	5615617	3	4	FLG BLIND 2500# RF LTCS A350 LF2 CL.1	0	0	
830	U04		BLIND	5615615	0.75	2	FLG BLIND 2500# RTJ LTCS A350 LF2 CL.1	0	0	
840	U04	817	FLANGE	5619926	3	3	FLG COMPACT 2500# RJ LTCS A350 LF2 CL.1 WT 16.0MM LINE CLASS 2500K1	0	0	
850	U04	817	FLANGE	5619904	6	6	FLG COMPACT 2500# RJ LTCS A350 LF2 CL.1 WT 29.0MM LINE CLASS 2500K1	0	0	
860	U04	817	BLIND	5619796	3	4	FLG COMPACT BLIND 2500# RJ LTCS A350 LF2 CL.1 LINE CLASS 2500K1	0	0	
870										
870							***** PIPE NIPPLES *****			
880										
880		812		0			NIPPLE BBE CUT FROM PIPE			
890										
890							***** GASKETS *****			
900										
900			GASKET	5672271	0.5	4	GASKET 2500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
910	U04		GASKET	5674480	0.5	2	GASKET 2500# RTJ SS OCTAGONAL RING	0	0	
920	U04	817	GASKET	5673521	3	6	GASKET 2500# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE LINE CLASS 2500K1	0	0	
930										
930		B1					***** BOLTING *****			
940										
940	U04		BOLTING	5675770	0.5	6	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
950										
950		SB6	BOLTING	5675675	12	12	JACK SCREW HEX HEAD A320 GR L7M	0	0	
960										
960							***** BLINDS & SPACERS *****			
970										
970			SPEC BLIND	5749597	3	3	SPECTACLE BLIND 2500# RF LTCS A516-GR70	0	0	
980			SPEC BLIND	5749590	0.75	2	SPECTACLE BLIND 2500# RTJ LTCS A516-GR70	0	0	
990	U04	817	SPEC BLIND	5749575	3	4	SPECTACLE BLIND COMPACT 2500# RJ LTCS A516-GR70	0	0	
1000										

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